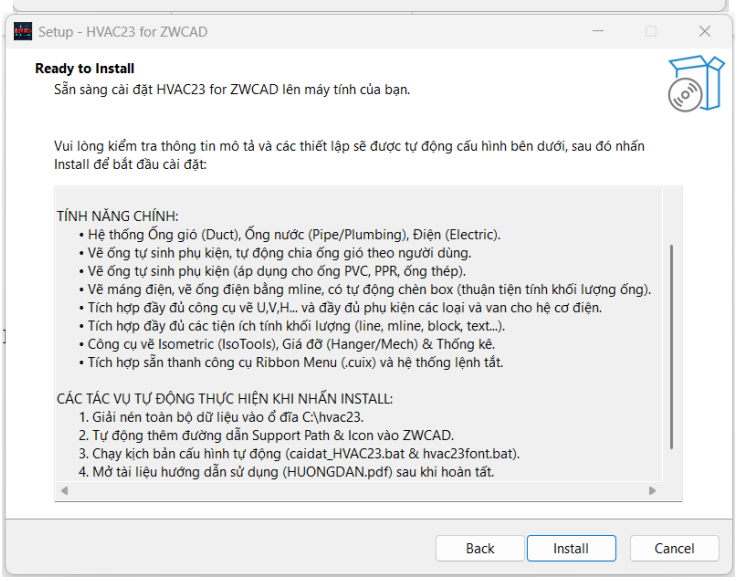
HƯỚNG DẪN CÀI ĐẶT VÀ SỬ DỤNG LISP

Lisp này mình tổng hợp từ lisp ishopDWG, HVAC, anh MAC, Trên cadviet và một số lisp khác...trong đó đã chỉnh sửa và thêm một số cho phù hợp. Lisp này có đủ công cụ để các bạn vẽ trong HVAC, ống nước... Trong quá trình sử dụng có thể gây ra lỗi. vì vậy rất mong các bạn chỉnh sửa hoạc góp ý để xây dựng được một lisp hỗ trợ vẽ trong HVAC hoàn thiện và đầy đủ hơn.

Chú ý: bộ HVAC23 chứa rất nhiều lisp nên khuyến khích các bạn không nên đổi lệnh mặc định của cad (vì có thể gây ra trùng lệnh)

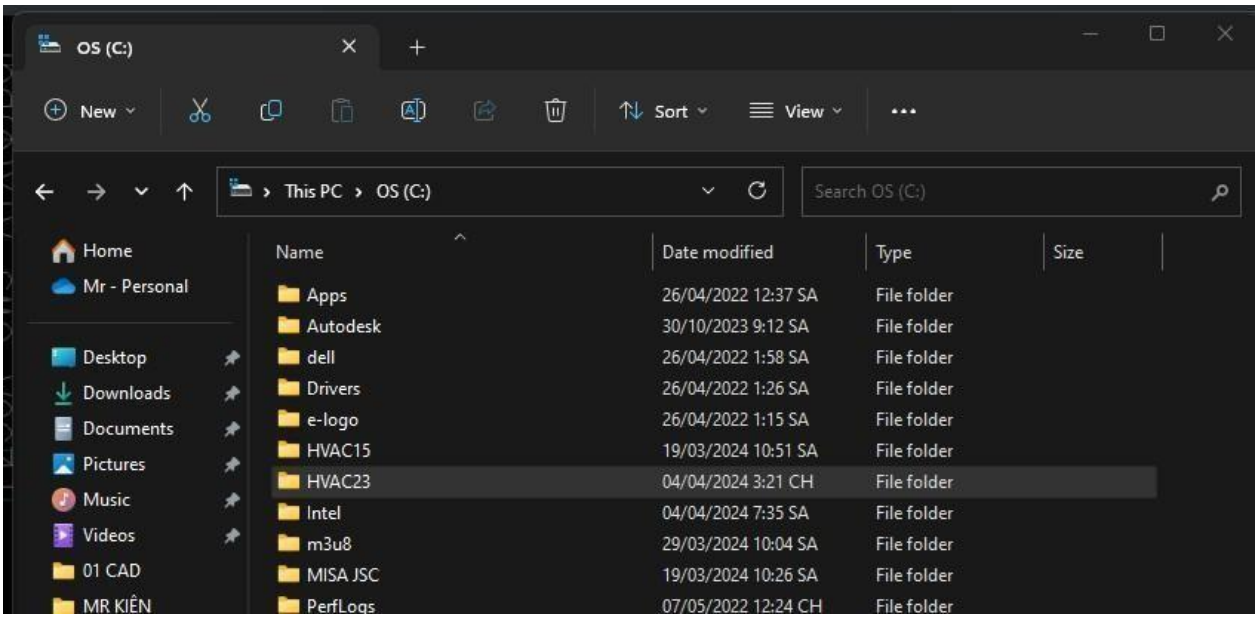
Step.1: Install the program





Bản quyền: free
Code: ISHOPDWG , HVAC, LEEMAC, Trên cadviet và nhiều nguồn khác nhau.
Chỉnh sửa: lê ngọc bách. (mọi ý kiến góp ý để hoàn thiện hơn xin gửi về email: ngocbach@gmail.com)

HƯỚNG DẪN CÀI ĐẶT VÀ SỬ DỤNG LISP HVAC23

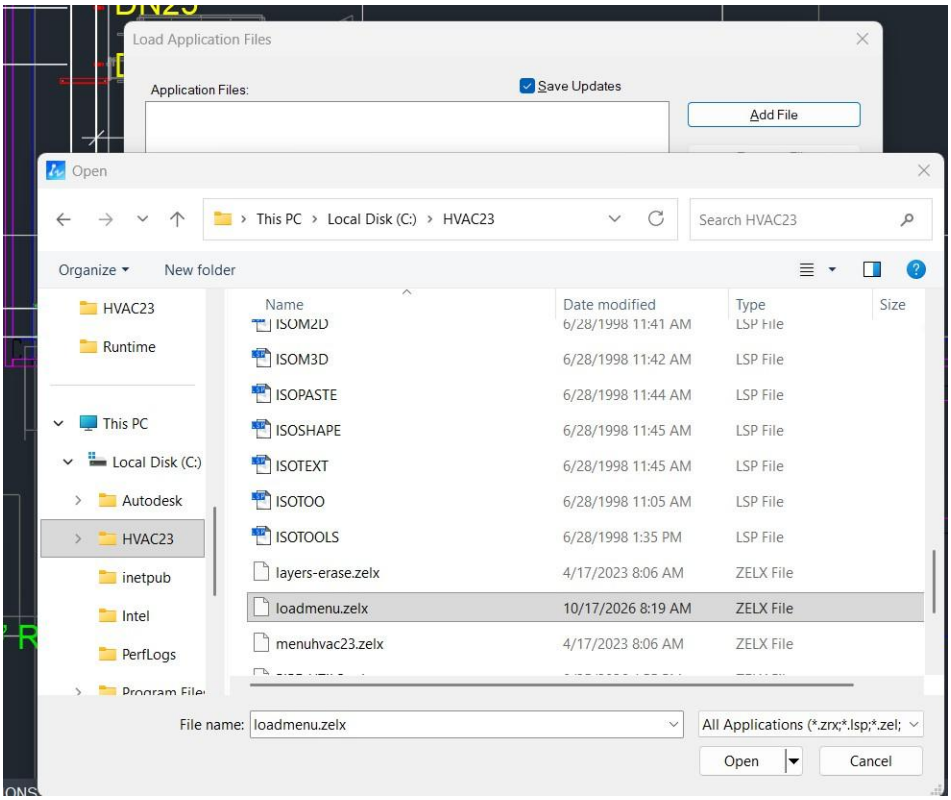


The HVAC23 folder must be located under the “C:/” folder.

Step.2: Open ZWCAD; if ZWCAD automatically loads HVAC23, you can skip this step.

- **Loading file “Loadmenu. Zelx” files**

Enter “Appload” in the command input window.



- **Sau đó nhấn load để Zwcad tự động thêm file parth và load ribbon tab menu.**

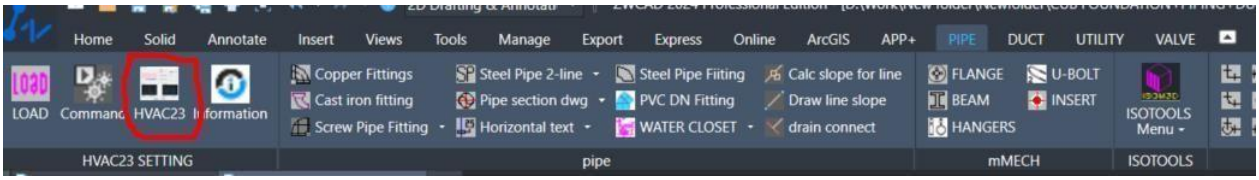
Bản quyền: free
Code: ISHOPDWG , HVAC, LEEMAC, Trên cadviet và nhiều nguồn khác nhau.
Chỉnh sửa: lê ngọc bách. (mọi ý kiến góp ý để hoàn thiện hơn xin gửi về email: ngocbach@gmail.com)

HƯỚNG DẪN CÀI ĐẶT VÀ SỬ DỤNG LISP HVAC23

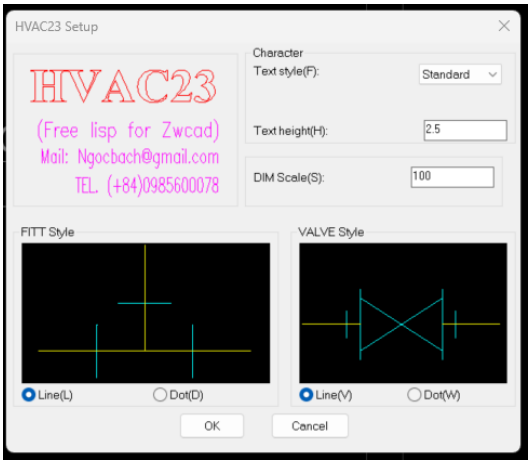
Verify that the “HVAC23” ribbon tab menu is displayed in the ribbon menu.



Exit and restart ZWcad.
Click “HVAC23 Settings” in the PIPE tab menu. Or type “HVAC23setup”.



Verify that the HVAC23 Setup dialog box appears. If a dialog box appears as shown below, you can use it normally.



If the HVAC23 Setup dialog box does not appear, proceed as follows.
If the above dialog box is displayed, you do not need to proceed with the steps below.

Exit and restart Zwcad.

Please refer to ngocbach@gmail.com for HVAC23 commands and usage instructions .

Bản quyền: free
Code: ISHOPDWG , HVAC, LEEMAC, Trên cadviet và nhiều nguồn khác nhau.
Chỉnh sửa: lê ngọc bách. (mọi ý kiến góp ý để hoàn thiện hơn xin gửi về email: ngocbach@gmail.com)

HƯỚNG DẪN CÀI ĐẶT VÀ SỬ DỤNG LISP HVAC23

HOW TO USE COMMAN HVAC23

ICON	command	explanation
	hvacsetup	Environment settings such as scale and character height, fitting shape, valve shape, etc.
	QQ1	Create 2-LINE with steel pipe outer diameter (Automatic insert fitting)
	QQ2	Create 2-LINE with PVC PIPE outer diameter (Automatic insert fitting)
	QQ3	Create 2-LINE with PVC PIPE outer diameter (Automatic insert fitting)
	CPIPE	Create 2-LINE using the outer diameter of the copper pipe
	IPIPE	Create 2-LINE with cast iron pipe outer diameter
	HD	Create horizontal pipe sizes
ICON	command	explanation
	CEO	Create vertical pipe sizes
	LD	Create Duct Size
	SWP	Insert piping name into piping line
	INT	Cut at random intervals when pipe lines cross each other
	SMO or Smove	Select and move elements
	SCP or Scopy	Select and copy elements
	SER or Serase	Select and delete elements
	SCH or Exchange	Select and change elements

Bản quyền: free
Code: ISHOPDWG , HVAC, LEEMAC, Trên cadviet và nhiều nguồn khác nhau.
Chỉnh sửa: lê ngọc bách. (mọi ý kiến góp ý để hoàn thiện hơn xin gửi về email: ngocbach@gmail.com)

HƯỚNG DẪN CÀI ĐẶT VÀ SỬ DỤNG LISP HVAC23

	**DUCT**	**Create a Duct Risen Section**
	**DUEL**	**Create a Duct Bend Fitting**
	**DUYD**	**Create a duct pant duct fitting**
	**DUBR**	**Create a tap fitting among the fittings connected to the main duct and branch duct.**
	**VVD**	**Create a Duct Damper**
	**FIT3**	**INSERT BLOCK PVC FITTING**
	**FIT1**	**INSERT BLOCK STEEL FITTING WELD**
	**EE3**	**INSERT BLOCK PPR FITTING**
	**FIT2**	**INSERT BLOCK SCREW STEEL FITTING**
ICON	**command**	**explanation**
	**FIT7**	**INSERT BLOCK HDPE FITTING SOCKET**
	**FIT6**	**INSERT BLOCK HDPE FITTING WELD**
	**SPR**	**INSERT BLOCK FITTING SPIRAL DUCT**
	**UBOLT**	**INSERT BLOCK UBOLT**
	**HANGER**	**INSERT BLOCK HANGER**
	**TML**	**TOTAL MLINE LENGTH**
	**DUD**	**Create duct up/down areas**
	**DCAL**	**Duct size calculation and size suggestions**

Bản quyền: free
Code: ISHOPDWG , HVAC, LEEMAC, Trên cadviet và nhiều nguồn khác nhau.
Chỉnh sửa: lê ngọc bách. (mọi ý kiến góp ý để hoàn thiện hơn xin gửi về email: ngocbach@gmail.com)

HƯỚNG DẪN CÀI ĐẶT VÀ SỬ DỤNG LISP HVAC23

	**DEND**	**Create Duct Ends**
	**HARD**	**Create a Duct Reducer Fitting**
	**SEC**	**Create a Pipe Section**
	**CUT**	**Create arbitrary pipe cutting areas**
	**FITT**	**Create steel pipe fittings (ELBOW, TEE, REDUCER, CAP, etc.)**
	**CFIT**	**Create copper pipe fittings (ELBOW, TEE, REDUCER, CAP, etc.)**
	**FLG**	**Written by FLANGE**
	**BEAM**	**Creating section steel (ANGLE, CHANNEL, H-BEAM, I-BEM, etc.)**
ICON	**command**	**explanation**
	**ELB**	**ELBOW symbol**
	**ELB1**	**ELB1**
	**EDN**	**ELBOW DOWN symbol**
	**THEM**	**ELBOW DOWN symbol**
	**ELL**	**ELBOW+ELBOW symbol**
	**TEE**	**TEE symbol**
	**TUP**	**TEE UP symbol**
	**TDN**	**TEE DOWN symbol**

Bản quyền: free
Code: ISHOPDWG , HVAC, LEEMAC, Trên cadviet và nhiều nguồn khác nhau.
Chỉnh sửa: lê ngọc bách. (mọi ý kiến góp ý để hoàn thiện hơn xin gửi về email: ngocbach@gmail.com)

HƯỚNG DẪN CÀI ĐẶT VÀ SỬ DỤNG LISP HVAC23

	**TTN**	**CROSS TEE symbol**
	**TSP**	**TEE + sprinkler head (bottom-up)**
	**RED**	**REDUCER symbol**
	**CAP**	**CAP symbol**
	**VAL**	**GATE VALVE symbol**
	**CHOICE1**	**GATE VALVE symbol of vertical piping**
	**VAL2**	**VALVE+STRAINER symbol of vertical piping**
	**VAL3**	**VALVE+CHECK VALVE symbol of vertical piping**
	**GVA**	**GLOBE VALVE symbol**
ICON	**command**	**explanation**
	**TWO**	**Diagram valve symbol**
	**MVA**	**MOTOR VALVE, constant flow valve symbol**
	**EVA**	**Solenoid valve symbol**
	**RVA**	**Pressure reducing valve symbol**
	**CHK**	**CHECK VALVE symbol**
	**STR**	**STRAINER Symbol**
	**ICO**	**Cast iron pipe cleaning tool symbol**
	**IYT**	**Cast iron pipe YT pipe symbol**

Bản quyền: free
Code: ISHOPDWG , HVAC, LEEMAC, Trên cadviet và nhiều nguồn khác nhau.
Chỉnh sửa: lê ngọc bách. (mọi ý kiến góp ý để hoàn thiện hơn xin gửi về email: ngocbach@gmail.com)

	**FLE**	**FLEXIBLE JOINT symbol**
	**RIS**	**pipe figurine symbol**
	**REF**	**relief valve symbol**
	**BALL**	**ball valve symbol**
	**BUTT**	**Butterfly valve symbol**
	**WHC**	**water hammer arrestor symbol**
	**HIM**	**union symbol**

HOW TO USE HVAC23

Before using HVAC23, first configure the environment.
Pipe fittings, copper pipe fittings, flanges, ducts, section steel, etc. are created according to other standards or data, but the size of pipe symbols or dimensioning text varies depending on the drawing scale.
In HVAC23, symbols, character sizes, etc. vary depending on the “ISCALE” value.

Bản quyền: free
Code: ISHOPDWG , HVAC, LEEMAC, Trên cadviet và nhiều nguồn khác nhau.
Chỉnh sửa: lê ngọc bách. (mọi ý kiến góp ý để hoàn thiện hơn xin gửi về email: ngocbach@gmail.com)

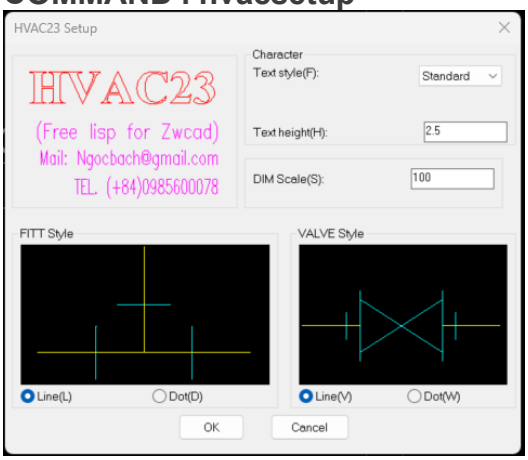
HƯỚNG DẪN CÀI ĐẶT VÀ SỬ DỤNG LISP HVAC23

ISCALE can be set by entering ISCALE in Command or in “Scale” in “HVAC23SETUP”. (ISCALE=Scale of Setting)

HVAC23 settings

To configure HVAC23, click “HVAC23” on the ribbon tab menu or enter “SETTING” in the command as shown below.

COMMAND : hvacsetup



- **Text style**

  This is a text type used for writing piping names, duct sizes, and pipe sizes. The initial value displays the character type of Styls.

- **Text height**

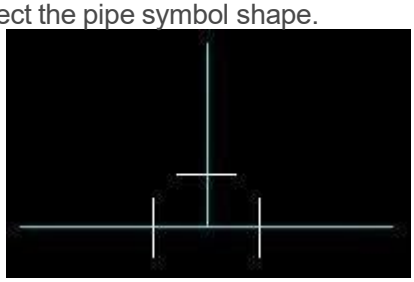
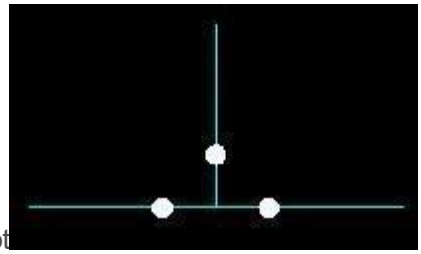
  This is the text size used to write piping names, duct sizes, and pipe sizes. The text size referred to here is the text size printed when the drawing is printed. The initial value is 2.5mm.

- **Dim scale**

  Drawing scale. (=ISCALE)

- **FITT style**

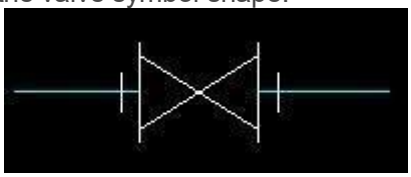
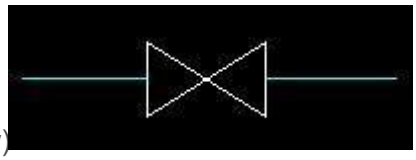
  Select the pipe symbol shape.

  line  dot 

- **VALVE style**

Bản quyền: free
Code: ISHOPDWG , HVAC, LEEMAC, Trên cadviet và nhiều nguồn khác nhau.
Chỉnh sửa: lê ngọc bách. (mọi ý kiến góp ý để hoàn thiện hơn xin gửi về email: ngocbach@gmail.com)

Select the valve symbol shape.

Line(v)  dot(w) 

For example, when the text shape is “GHS”, the text height is “2.5″, and the scale is “100″, the text size written in the drawing is written in a size of 2.5×100=250mm. Piping symbols or valve symbols increase or decrease in size depending on the scale. The drawing scale is “1/100″, but if you set the scale to “1”, the piping symbols and piping names will be written too small.

ISCALE

The drawing scale can be entered separately using “ISCALE” .

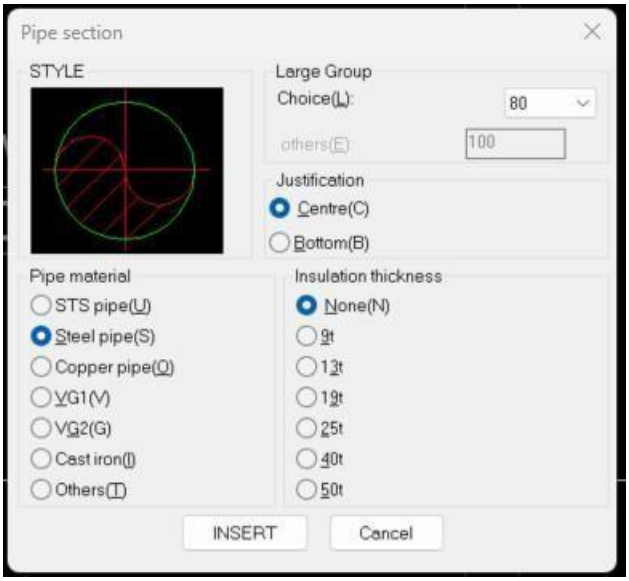
Command: ISCALE

Enter the Shop drawing scale<50>:100

SEC

This is a program that draws a Taegeuk-shaped pipe cross section.
You can create by selecting the type such as steel pipe, copper pipe, PVC PIPE, etc.
This function is used using blocks, so when you want to change the layer or color, open the C-SEC.DWG and B-SEC.DWG files in the “PIPE” folder and modify them.
You can randomize the size by clicking the “Other” option.

Command: SEC



CUT

Bản quyền: free
Code: ISHOPDWG , HVAC, LEEMAC, Trên cadviet và nhiều nguồn khác nhau.
Chỉnh sửa: lê ngọc bách. (mọi ý kiến góp ý để hoàn thiện hơn xin gửi về email: ngocbach@gmail.com)

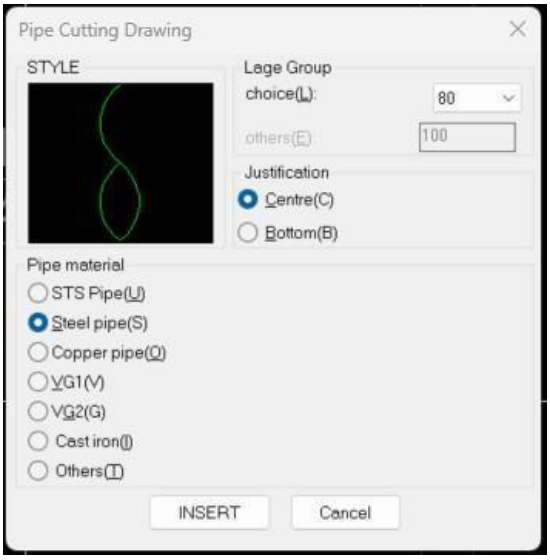
HƯỚNG DẪN CÀI ĐẶT VÀ SỬ DỤNG LISP HVAC23

This function creates a pipe cutting surface.

This function is used using blocks, so when you want to change the layer or color, open the C-CUT.DWG and B-CUT.DWG files in the “PIPE” folder and modify them.

You can randomize the size by clicking the “Other” option.

Command: CUT



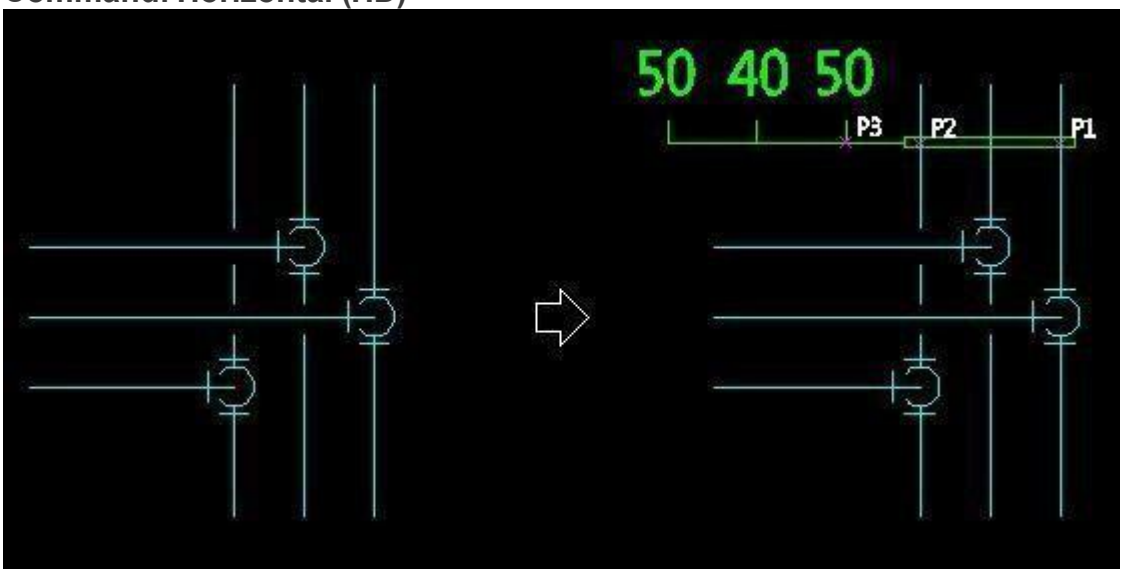
HD,VD

This is the function to create multiple pipe sizes.

When writing horizontally, enter :HD. When writing vertically, enter :VD.

The size and style of the text vary depending on the style, size, and scale values set in “Setting”.

Command: Horizontal (HD)



Loading.....——You can used command VD , HD

Bản quyền: free
Code: ISHOPDWG , HVAC, LEEMAC, Trên cadviet và nhiều nguồn khác nhau.
Chỉnh sửa: lê ngọc bách. (mọi ý kiến góp ý để hoàn thiện hơn xin gửi về email: ngocbach@gmail.com)

** Pipe dia dim. dwg**

First point :P1

Second point :P2

Text point :P3

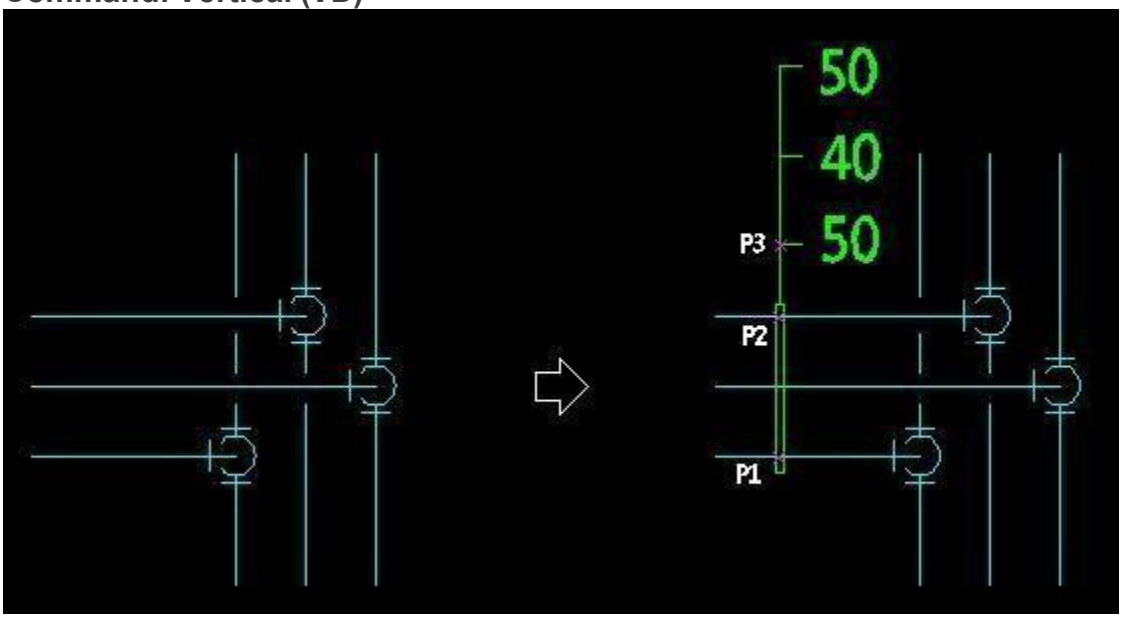
Enter the dimtxt:50

Enter the dimtxt:50

Enter the dimtxt:32

Enter the dimtxt:ENTER

Command: Vertical (VD)



Loading..... ——You can used command VD , HD

** Pipe dia dim. dwg**

First point :P1

Second point :P2

Text point :P3

Enter the dimtxt:20

Enter the dimtxt:20

Enter the dimtxt:20

Enter the dimtxt:ENTER

SPIPE , CPIPE , PPIPE , IPIPE, RPIPE, HPIPE

Bản quyền: free
Code: ISHOPDWG , HVAC, LEEMAC, Trên cadviet và nhiều nguồn khác nhau.
Chỉnh sửa: lê ngọc bách. (mọi ý kiến góp ý để hoàn thiện hơn xin gửi về email: ngocbach@gmail.com)

HƯỚNG DẪN CÀI ĐẶT VÀ SỬ DỤNG LISP HVAC23

This is the function to create a pipe as TWO LINE.

The commands are steel pipe: SPIPE, copper pipe: CPIPE, PVC PIPE:PPIPE, and cast iron pipe:IPIPE.

Comand: SPIPE

Loading ……. ——You can used command CPIPE,SPIPE,PPIPE,IPIPE,RPIPE,HPIPE

Steel pipe 3 lines dwg

Pipe diameter 15 20 25 32 40 50 65 80 100 125 150 200 250 300 350 400 450 500 550 600 650 700 750 800 850 900 950 1000 1050

Please enter the pipe diameter steel pipe<80>:100

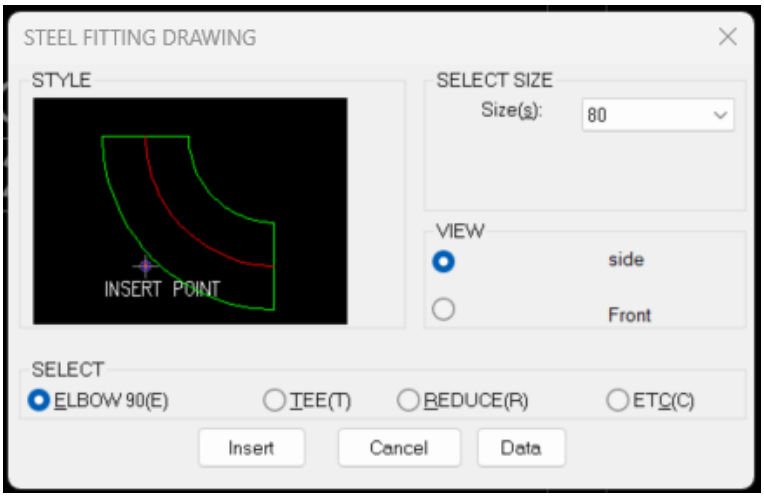
FITT

This is a function to create welded elbows, tees, reducers, caps, and 45-degree elbows among steel pipe fittings in TWO-LINE format.

Click the “Data” button to see a detailed dimension table for each fitting.

For reference, the size with “S” attached to the elbow is SHORT TYPE.

Command: FITT



FIT1

This is a function to insert dyamic block welded elbows, tees, reducers, caps, and 45-degree elbows among steel pipe fittings in TWO-LINE format.

Command: FIT1

Bản quyền: free
Code: ISHOPDWG , HVAC, LEEMAC, Trên cadviet và nhiều nguồn khác nhau.
Chỉnh sửa: lê ngọc bách. (mọi ý kiến góp ý để hoàn thiện hơn xin gửi về email: ngocbach@gmail.com)

HƯỚNG DẪN CÀI ĐẶT VÀ SỬ DỤNG LISP HVAC23



FIT2

This is a function to insert dyamic block screw elbows, tees, reducers, caps, and 45-degree elbows among steel pipe fittings in TWO-LINE format.

Command: FIT2



EE1

This is a function to insert dyamic block PVC elbows, flange, CO, and 45-degree elbows among PVC pipe fittings in TWO-LINE format.

Command: EE1



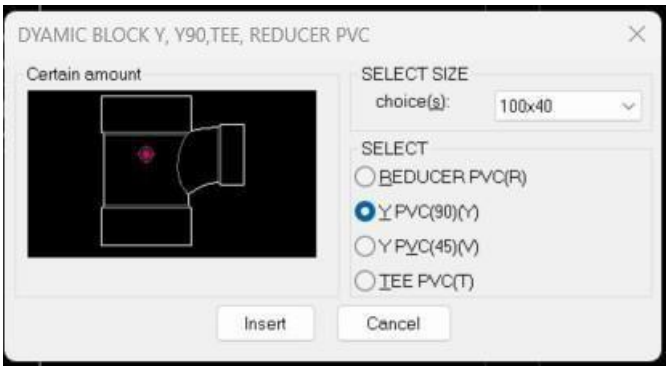
EE2

Bản quyền: free
Code: ISHOPDWG , HVAC, LEEMAC, Trên cadviet và nhiều nguồn khác nhau.
Chỉnh sửa: lê ngọc bách. (mọi ý kiến góp ý để hoàn thiện hơn xin gửi về email: ngocbach@gmail.com)

HƯỚNG DẪN CÀI ĐẶT VÀ SỬ DỤNG LISP HVAC23

This is a function to insert dyamic block PVC reducer, y, y90, and tee pipe fittings in TWO-LINE format.

Command: EE2



EE3

This is a function to insert dyamic block PPR elbows, vale, reducer, tee, and 45-degree elbows among PPR pipe fittings in TWO-LINE format.

Command: EE3



FIT3

This is a function to insert dyamic block PVC elbows, Y, Y90, TEE, CO, and p-trap among PVC pipe fittings in TWO-LINE format.

Command: FIT3

Bản quyền: free
Code: ISHOPDWG , HVAC, LEEMAC, Trên cadviet và nhiều nguồn khác nhau.
Chỉnh sửa: lê ngọc bách. (mọi ý kiến góp ý để hoàn thiện hơn xin gửi về email: ngocbach@gmail.com)

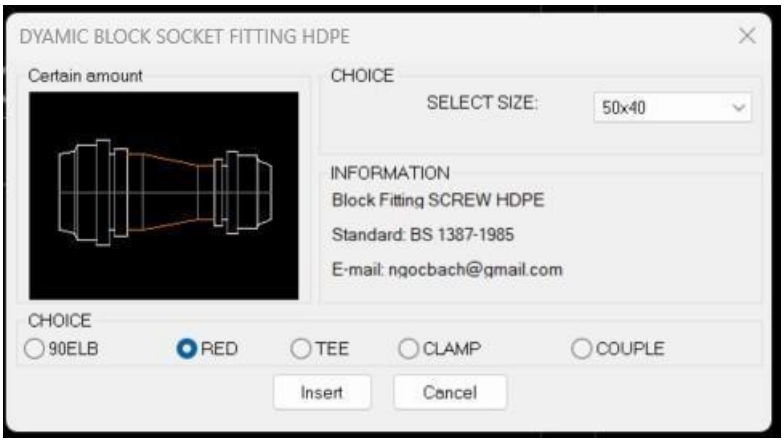
HƯỚNG DẪN CÀI ĐẶT VÀ SỬ DỤNG LISP HVAC23



FIT7

This is a function to insert dyamic block socket HDPE elbows, reducer,TEE, clamp, and couple among HDPE pipe fittings in TWO-LINE format.

Command: FIT7



FIT6

This is a function to insert dyamic block welder HDPE elbows, reducer,TEE, clamp, and couple among HDPE pipe fittings in TWO-LINE format.

Command: FIT6



BEAM

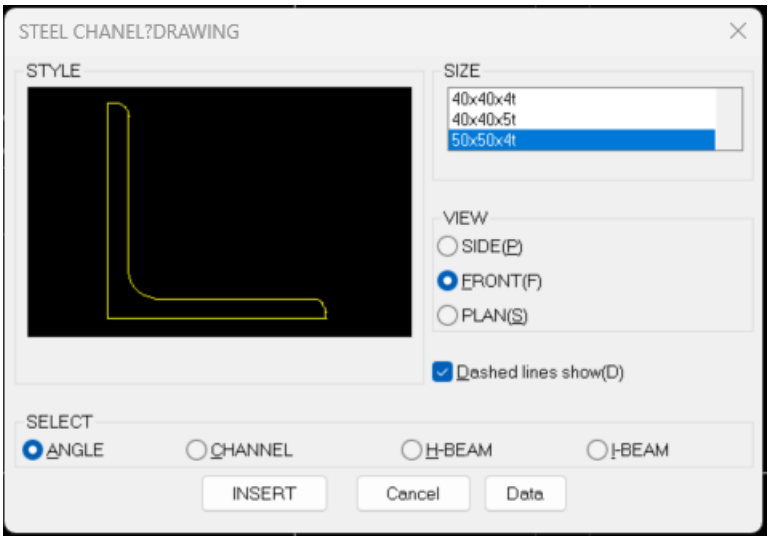
Bản quyền: free
Code: ISHOPDWG , HVAC, LEEMAC, Trên cadviet và nhiều nguồn khác nhau.
Chỉnh sửa: lê ngọc bách. (mọi ý kiến góp ý để hoàn thiện hơn xin gửi về email: ngocbach@gmail.com)

HƯỚNG DẪN CÀI ĐẶT VÀ SỬ DỤNG LISP HVAC23

This is the function to create Angle, Channel, H-Beam, and I-Beam by selecting the specifications from the list.

You can create a front view, side view, etc.

Command: BEAM



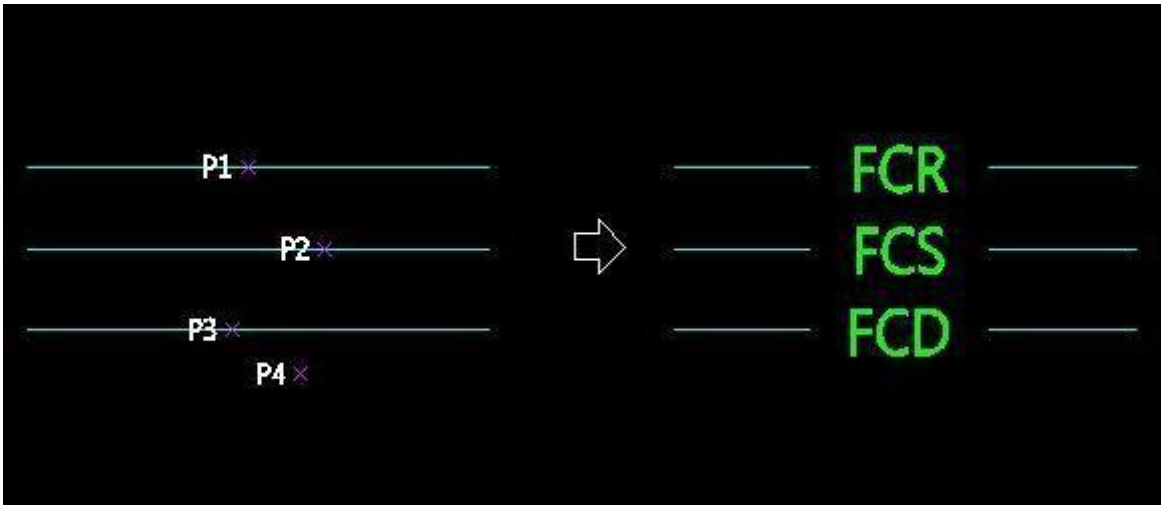
SWP

When entering a piping name at a random location on a piping line, this function is used to enter the piping name on multiple piping lines and at the same time cut the piping lines at regular intervals.

Select the pipe lines in order and enter the pipe names in the selected order.

The size and style of the text vary depending on the style, size, and scale values set in “Setting”.

Command: SWP



>>Pipe symble write Program…..

Select objects: P1

Select objects: P2

Bản quyền: free
Code: ISHOPDWG , HVAC, LEEMAC, Trên cadviet và nhiều nguồn khác nhau.
Chỉnh sửa: lê ngọc bách. (mọi ý kiến góp ý để hoàn thiện hơn xin gửi về email: ngocbach@gmail.com)

Select objects: P3

Select objects:Enter

Pipe Symble Write point:P4

Enter Pipe symble: FCS

Enter Pipe symble: FCD

Enter Pipe symble: FCR

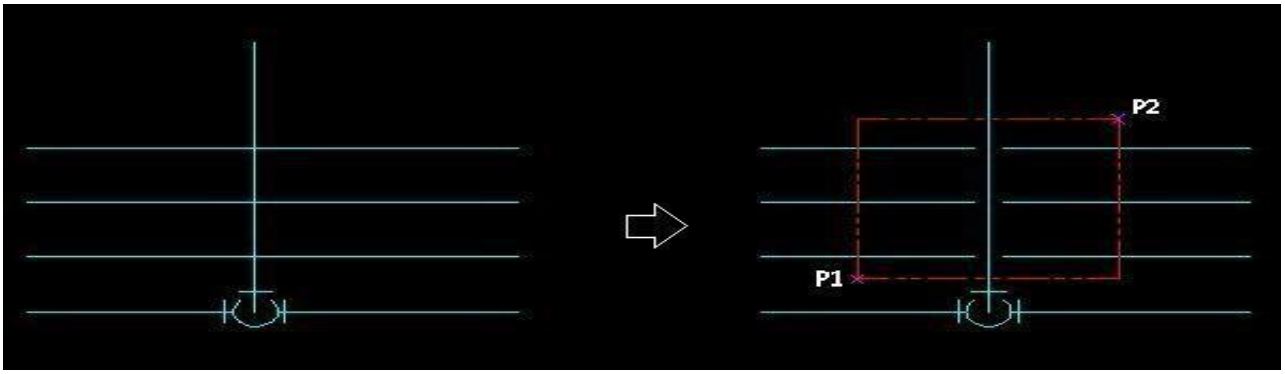
INT

This function cuts the area where horizontal and vertical lines intersect in a schematic diagram by cutting it to an arbitrary size.

Set the desired size in the horizontal and vertical directions and automatically cut out the intersecting areas.

The cut size varies depending on the Dist size and the scale value of ISCALE or Setting.

Command: INT



INT.LSP loaded. Start command with INT.

>>Interstion point multe trim Program…..

Select objects: C

First corner: Other corner: 7 found

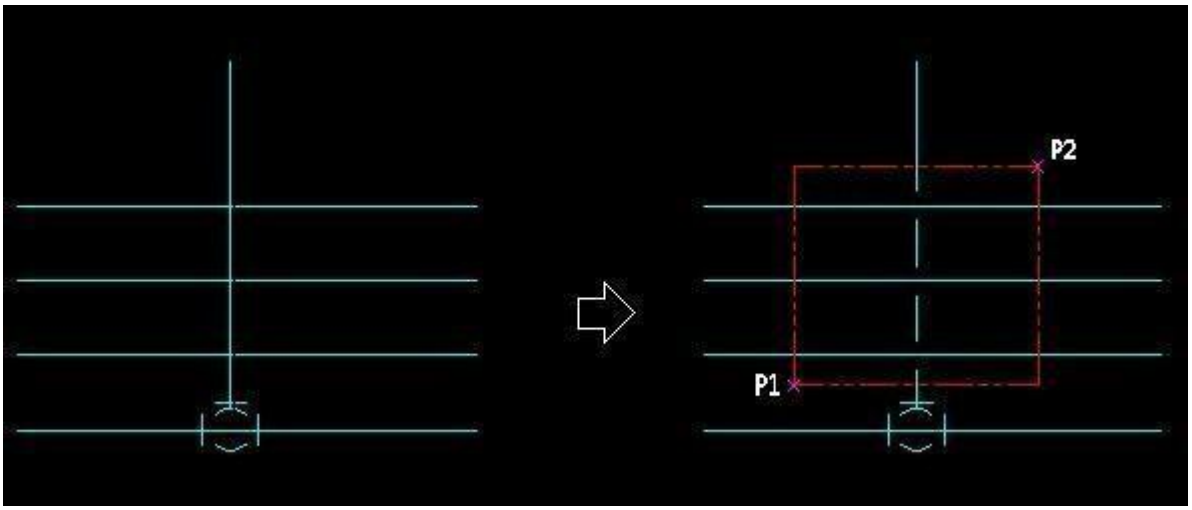
Select objects:ENTER

Select Horizntl/Vertical/Dist: H

Command: INT

Bản quyền: free
Code: ISHOPDWG , HVAC, LEEMAC, Trên cadviet và nhiều nguồn khác nhau.
Chỉnh sửa: lê ngọc bách. (mọi ý kiến góp ý để hoàn thiện hơn xin gửi về email: ngocbach@gmail.com)

HƯỚNG DẪN CÀI ĐẶT VÀ SỬ DỤNG LISP HVAC23



INT.LSP loaded. Start command with INT.

>>Interstion point multe trim Program…..

Select objects: C

First corner: Other corner: 7 found

Select objects:ENTER

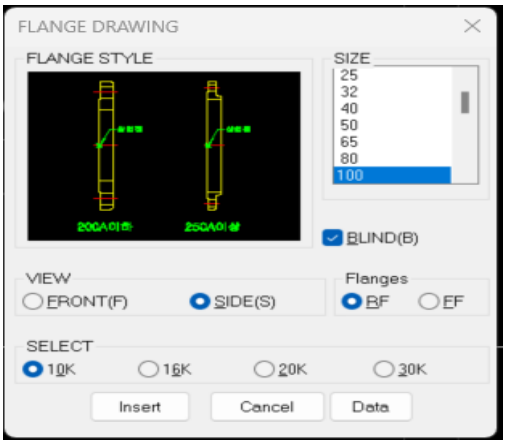
Select Horizntl/Vertical/Dist: V

FLG

This is the function to create FLANGE by plan view, side view, and pressure.

You can mark or unmark bolt holes as needed.

Command: FLG



CFIT

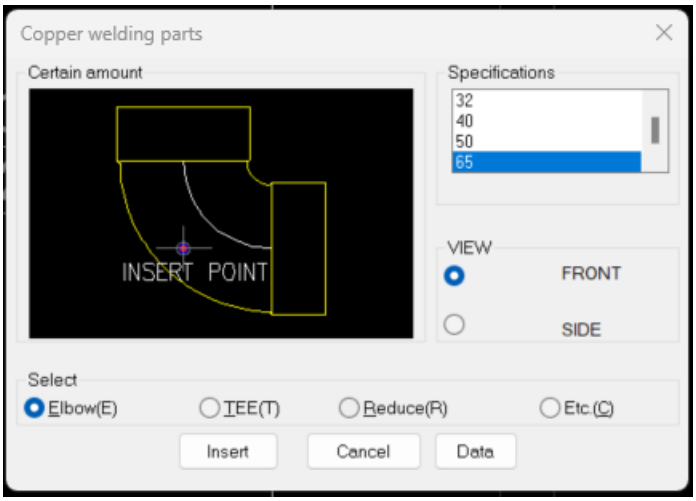
This function creates copper pipe fittings.

You can create elbow, tee, reducer, cap, and 45 degree elbow fittings, and you can choose the front view or top view shape to create them.

Command: CFITT

Bản quyền: free
Code: ISHOPDWG , HVAC, LEEMAC, Trên cadviet và nhiều nguồn khác nhau.
Chỉnh sửa: lê ngọc bách. (mọi ý kiến góp ý để hoàn thiện hơn xin gửi về email: ngocbach@gmail.com)

HƯỚNG DẪN CÀI ĐẶT VÀ SỬ DỤNG LISP HVAC23



ELB,ELB1,EDN,TEE,ELL,TUP,TDN,TTN,TSP,CAP,RED,STR,VAL

This function creates pipe symbols.

You can create symbols for major fittings and valves in your pipes.

This function does not use CAD blocks and is displayed as a new one created in Lisp, and unnecessary lines inside the symbol are automatically cut out.

The size of the symbol varies depending on the Iscale value.

Change the Iscale value by entering “Iscale” in the command, or enter “Setting” to change the “scale” in the ISHOP settings dialog box. (Scale of Setting = ISCALE)



∗ When creating symbols, OSNAP mode is set by default. It is not done.

If you want to accurately mark the pipe line, enter the Osnap mode command and click the insertion point.

LD

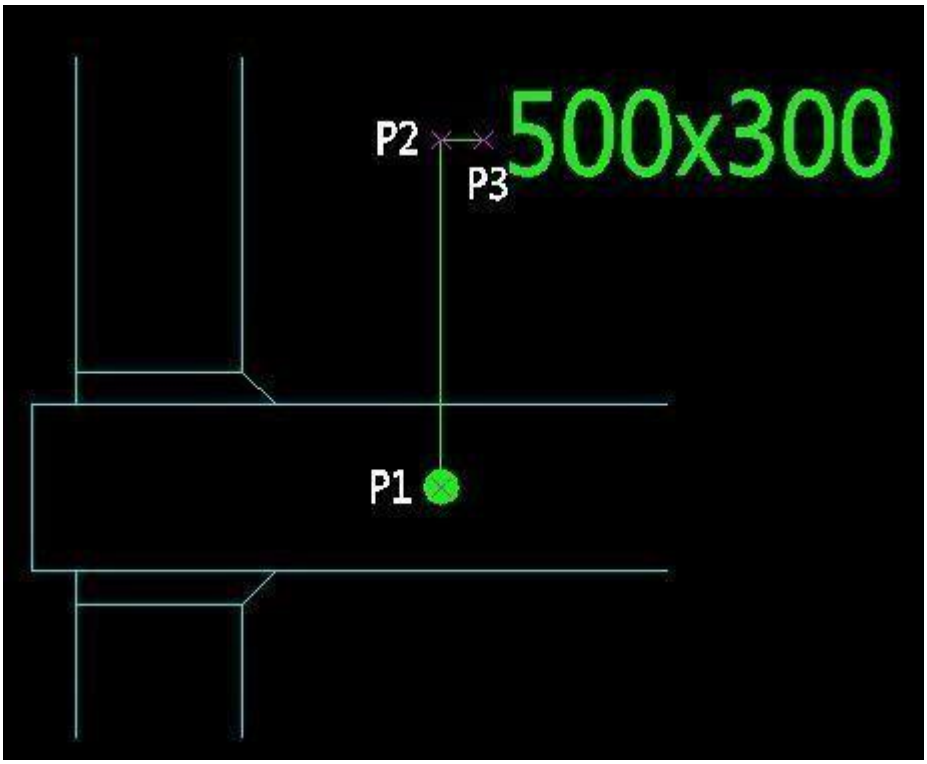
Ability to create duct sizes.

The size, style, and dot size of the text vary depending on the style, size, and scale values set in “Setting”.

Command: LD

Bản quyền: free
Code: ISHOPDWG , HVAC, LEEMAC, Trên cadviet và nhiều nguồn khác nhau.
Chỉnh sửa: lê ngọc bách. (mọi ý kiến góp ý để hoàn thiện hơn xin gửi về email: ngocbach@gmail.com)

HƯỚNG DẪN CÀI ĐẶT VÀ SỬ DỤNG LISP HVAC23



Leader start point :P1

To point :P2

Duct size angle :P3

Enter the duct size:500

Enter the duct size:300

Command: LD



Leader start point :P1

To point :P2

Bản quyền: free
Code: ISHOPDWG , HVAC, LEEMAC, Trên cadviet và nhiều nguồn khác nhau.
Chỉnh sửa: lê ngọc bách. (mọi ý kiến góp ý để hoàn thiện hơn xin gửi về email: ngocbach@gmail.com)

HƯỚNG DẪN CÀI ĐẶT VÀ SỬ DỤNG LISP HVAC23

Duct size angle :P3

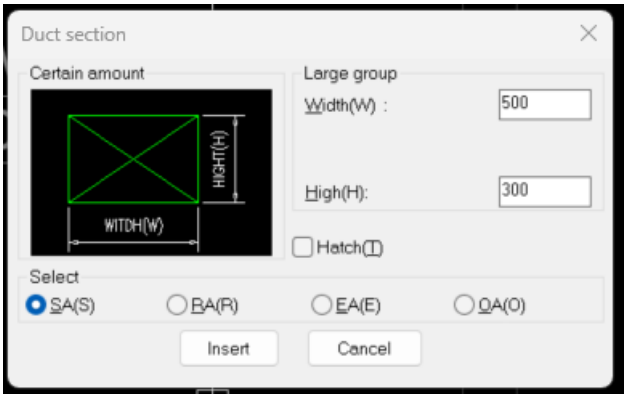
Enter the duct size:400

Enter the duct size:250

DUCT

This is a function to create the granular cross-sectional shape of the duct (air supply, exhaust, ventilation, outside air).

Command: DUCT



DUEL

This function creates 90-degree, 45-degree, 30-degree, and 15-degree curved duct shapes. You can create duct bends with different shapes on the inlet and outlet sides. If the duct width is unknown, you can select a line to create it.

Command: DUEL



DUYD

This is a function to create branch fittings in the shape of pants (2-way, 3-way) among duct branching methods.

Command: DUYD

Bản quyền: free
Code: ISHOPDWG , HVAC, LEEMAC, Trên cadviet và nhiều nguồn khác nhau.
Chỉnh sửa: lê ngọc bách. (mọi ý kiến góp ý để hoàn thiện hơn xin gửi về email: ngocbach@gmail.com)

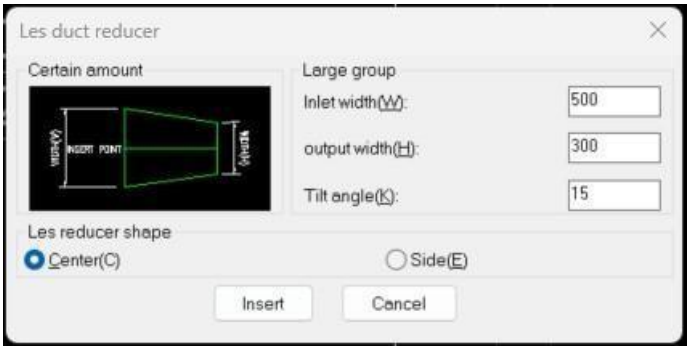
HƯỚNG DẪN CÀI ĐẶT VÀ SỬ DỤNG LISP HVAC23



DUBR

This is the function to create a tap fitting among the fittings connected to the main duct and branch duct of the duct.

Command: DUBR



VVD

Ability to create duct dampers (VD, MD, FVD).

You can select the shape of the duct damper, and if the size of the duct is unknown, you can select the duct line to create it.

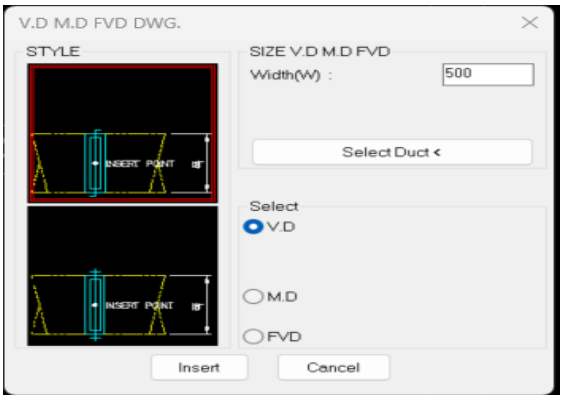
You can also write text indicating the type of duct wherever you want.

The size and style of the text are created according to the style and size set in “Setting“.

Command: VVD

Bản quyền: free
Code: ISHOPDWG , HVAC, LEEMAC, Trên cadviet và nhiều nguồn khác nhau.
Chỉnh sửa: lê ngọc bách. (mọi ý kiến góp ý để hoàn thiện hơn xin gửi về email: ngocbach@gmail.com)

HƯỚNG DẪN CÀI ĐẶT VÀ SỬ DỤNG LISP HVAC23



DUUD

This is a function to create the UP-DOWN part of the duct.

It is created in plan and cross-section form, and you can also select the up-down type.

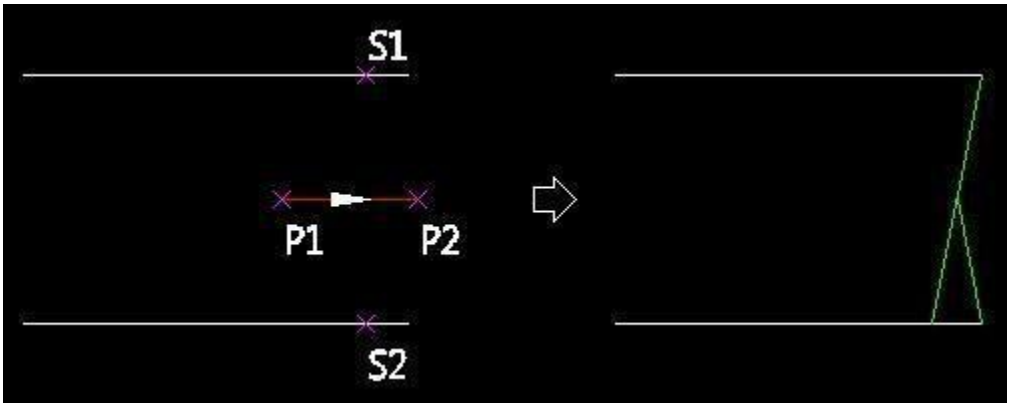
Command: DUUD



DEND

This is the function to create the end part of the duct.

Command: DEND



Loading DEND.lsp——You can used commnad DEND

Select a duct line?: S1

Bản quyền: free
Code: ISHOPDWG , HVAC, LEEMAC, Trên cadviet và nhiều nguồn khác nhau.
Chỉnh sửa: lê ngọc bách. (mọi ý kiến góp ý để hoàn thiện hơn xin gửi về email: ngocbach@gmail.com)

Select a different duct line?: S2

Where does the duct end?: P1

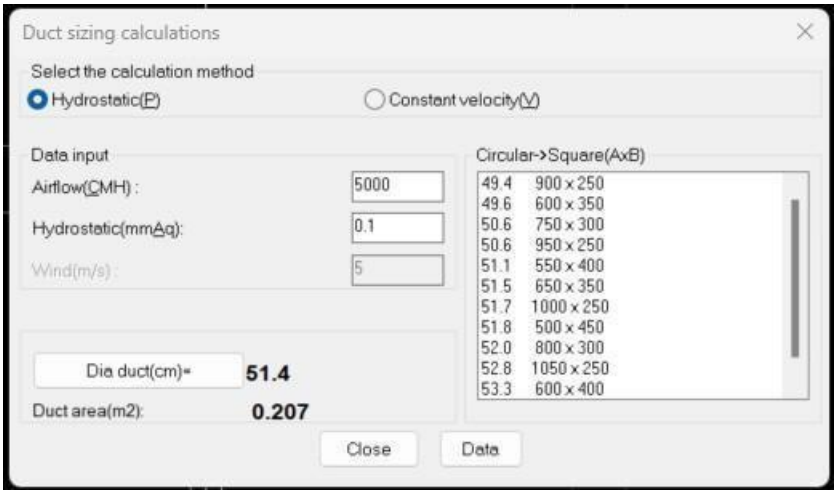
In what direction is the duct cut?: P2

DCAL

This function selects the size of the duct using the static pressure method and constant velocity method by inputting the air volume value of the duct.

It is calculated as a circular duct and is displayed in a list so that you can select the square duct size.

Command: DCAL



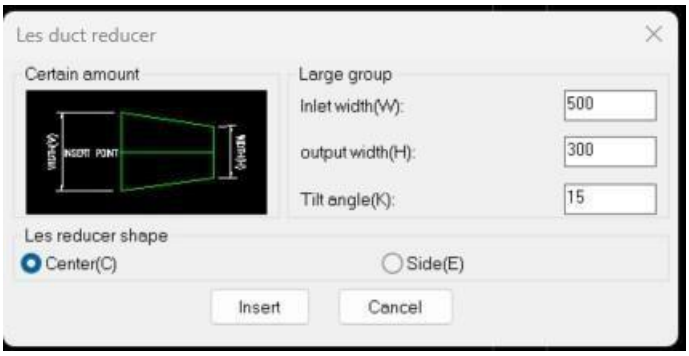
Click the “Data” button to view data related to duct calculations.

HARD

This is the function to create a doctor reducer.

You can select a concentric or eccentric shape and create it by entering the angle of the reducer.

Command: DURE



SPIRAL

This is the function to create a elbow, tee, reducer.

Bản quyền: free
Code: ISHOPDWG , HVAC, LEEMAC, Trên cadviet và nhiều nguồn khác nhau.
Chỉnh sửa: lê ngọc bách. (mọi ý kiến góp ý để hoàn thiện hơn xin gửi về email: ngocbach@gmail.com)

HƯỚNG DẪN CÀI ĐẶT VÀ SỬ DỤNG LISP HVAC23

Command: SPR



SERASE,SMOVE,SCOPY,SCHANGE

- **SERASE**

  This function allows you to selectively delete only what you want when editing complex drawings. You can select and delete only the layers, colors, line types, entities, etc. you want.

- **SMOVE**

  When editing complex drawings, you can selectively move only what you want. You can select and move only the layers, colors, line types, entities, etc. you want.

- **SCOPY**

  This function allows you to selectively copy only what you want when editing complex drawings. You can select and copy only what you want, such as layers, colors, line types, and entities.

- **CHANGE**

  This function allows you to selectively change only what you want when editing complex drawings. You can select and change only the layer, color, line type, entity, etc. you want.

AND OTHER COMMAND…

Bản quyền: free
Code: ISHOPDWG , HVAC, LEEMAC, Trên cadviet và nhiều nguồn khác nhau.
Chỉnh sửa: lê ngọc bách. (mọi ý kiến góp ý để hoàn thiện hơn xin gửi về email: ngocbach@gmail.com)